MIL-M-24225A(SH)
3 March 1987
SUPERSEDING
MIL-M-24225(SHIPS)
14 June 1966
(See 6.8)

MILITARY SPECIFICATION

MONITOR, FREQUENCY, 60 HERTZ, ELECTRICAL (NAVAL SHIPBOARD USE)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1 1 Scope. This specification covers frequency monitors for sensing a low frequency value on one phase of a three phase 60 hertz (Hz) circuit.

1 2 Classification. The monitor shall be of the following types as specified (see 6.2.1).

(a) Type FM60 - (bulkhead mounted), 52 Hz trip
(b) Type FM61 - (switchboard mounted), 54 to 57 Hz adjustable trip

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Test, H I (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5925
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-S-16036 - Switchgear, Power, Naval Shipboard.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts): Packaging of.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-701 - List of Standard Semiconductor Devices.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests· Exponential Distribution.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems Electric Power, Alternating Current (Metric).
MIL-STD-1562 - Lists of Standard Microcircuits.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Safety. The monitor shall provide fail-safe features for safety of personnel during the installation, operation, maintenance, and repair or interchanging of a complete monitor assembly or component parts thereof.

3.2 1 Accidental contact The construction of the monitor shall provide protection to personnel against accidental contact with voltages in excess of 30 volts root mean square (rms) alternating current (ac) or 30 volts direct current (dc).

3.2.2 Leakage current. Leakage current shall not exceed 5 milliamperes in accordance with MIL-E-917 when tested as specified in 4.8.3.

3 3 Materials, parts, and processes. Unless otherwise specified (see 6.2.1), materials, parts, and processes shall be in accordance with MIL-E-917.

3.3 1 Recovered materials Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3.2 Hazardous items and toxic materials. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred to the Naval Sea Systems Command Regardless of any other requirements, materials and parts containing asbestos, cadmium, lithium, magnesium based alloys, mercury, radioactive material, or materials giving off toxic fumes under operation or casualty conditions shall not be used.

3 3.3 Material compatibility. Materials which, through outgassing or other physical phenomena, cause deterioration of other materials or degradation of equipment performance shall not be used.

3.3 4 Nonstructural parts. Nonstructural parts such as terminal boxes, housings, ducts, and similar parts shall be fabricated from aluminum alloy, where feasible.

3.4 Construction Unless otherwise specified herein, the construction of the monitors shall be in accordance with MIL-E-917.

3.4.1 Shipboard system to be monitored. The shipboard system to be monitored shall be in accordance with DOD-STD-1399, section 300.

3.4.2 Monitor features. Monitor performance shall be in accordance with the following requirements.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T20:08Z
Check the source to verify that this is the current version before use.

(a) 440 volts.
(b) 60 ± 3 Hz.
(c) Single phase.
(d) Trip band:
FM60: Trip setting 52 Hz; in trip condition, 0-52 ± 1 Hz.
FM61: Trip setting 54 to 57 Hz minimum adjustable; in trip condition, 0 to setting plus or minus 1/2 Hz.
(e) Trip point time: 50 milliseconds (ms) or less.
(f) Trip point closing contact rating: 25 watts at 15 percent power factor.
(g) Reset: Automatic and in 50 ms or less.
(h) Warmup time: 1 minute.
(i) Design: Static.
(j) Enclosure:
FM60: Dripproof.
FM61. Totally enclosed.
(k) Ventilation: Natural.
(l) Ambient temperature:
FM60: 0 to 50 degrees Celsius (°C).
FM61: 0 to 65°C.

3.4.3 Enclosures. It shall be possible to remove or install any internal component without removing the enclosure from its mounting

3.4.3.1 Bulkhead mounted. Bulkhead mounted monitors shall be of dripproof protected construction to 15°C in accordance with MIL-E-2036 Mounting means shall protrude from the top and the bottom of the enclosure. The mounting holes shall be suitable to take 1/4-20 unified coarse (UNC) screws A wiring compartment 2 inches high shall be provided at the bottom of the enclosure and a cable knockout of 1-1/32 inch diameter shall be placed in the two sides of the enclosure. In the bottom of the enclosure, a 5/16-inch screw shall be provided as a possible condensation drain or as a breather hole (see 4.8 1)

3 4.3.2 Switchboard mounted. Switchboard mounted monitors shall be totally enclosed with the outline and mounting dimensions as shown on figure 1 to provide interchangeability of monitors between manufacturers

3.4.4 Mounting The FM60 type monitor shall be bulkhead mounted. The FM61 type monitor shall be mounted within a switchboard enclosure in accordance with MIL-S-16036.

3.4.5 Input power. The monitors shall meet all performance requirements specified herein when powered by type I power in accordance with DOD-STD-1399, section 300 (see 4.8.4).

3.4.6 Internal components. Components shall be accessible only when the front cover of the enclosure is removed Small static type components shall be mounted on a removable control board Unless otherwise prohibited (see 6.2.1), a printed circuit control board is permitted. An internal single throw control switch shall be provided for defeating the monitor trip circuit, if found necessary. All external wiring shall connect to an easily accessible screw type terminal block.

Source: https://assist.dla.mil -- Downloaded: 202640-19T20:08Z
Check the source to verify that this is the current version before use.

3.4.7 Solid state components. Semiconductors shall be in accordance with MIL-S-19500 with preferred sizes given in MIL-STD-701, and microcircuits shall be in accordance with MIL-M-38510 with preferred sizes given in MIL-STD-1562. Controlled rectifiers and diode rectifiers shall be of the silicon type in accordance with MIL-STD-701.

3.4 8 Duty The monitor shall sense the generator circuit continuously. The monitor shall meet the performance requirement of test condition 3 (see 4.8.14.3).

3.4.9 Operating voltage. Unless otherwise specified (see 6.2.1), the monitor shall pick up below 374 volts and drop out below 270 volts.

3.4.10 Operating frequency. Unless otherwise specified (see 6.2 1), the operating frequency shall be 60 ± 3 Hz and shall operate on type I power in accordance with DOD-STD-1399, section 300

3 5 Reliability. The upper test mean time between failure (MTBF) for continuous and intermittent operation in accordance with MIL-STD-781 shall be 15000 hours under the environmental conditions specified for the monitor when tested in accordance with 4.8.15. A failure (as used herein) shall be as specified in 4.8.18

3 6 Maintainability. The monitor shall facilitate trouble-shooting, fault isolation, and repair down to the lowest nonrepairable part or nonrepairable assembly (see 4.8.16).

3.7 Performance/sensing The monitor, when applied as shown on figure 2, shall sense for the low frequency band specified herein on one phase of a three phase shipboard system as specified in 3.4.1. The monitor shall maintain its trip contact closed until the sensed frequency exceeds the specified trip setting. The trip contact shall have a rating of 25 watts at 15 percent power factor. After the monitor has tripped the system circuit breaker, the monitor shall automatically reset to permit closing of the circuit breaker under favorable frequency conditions (above the specified trip band). The monitor shall not be damaged because of shorted or grounded sensing or breaker trip leads. A toggle type switch connected in series with the breaker trip circuit shall be provided within the monitor enclosure to permit defeat of the monitor if required. The semiconductor components shall be adequately protected against high transient voltage peaks which may occur. See DOD-STD-1399, section 300 for allowable limits on transient voltage peaks (see 4.8.14).

3.8 Environmental. Unless otherwise specified (see 6.2.1), the monitor shall be in accordance with the following environmental requirements:

3.8.1 Ambient temperature. The monitor shall meet all the specification requirements when operating in an ambient temperature from 0 to 65°C and have no more than 15°C temperature rise of the case (see 4.8.5).

3.8 2 Humidity. The monitor shall meet the requirements of this specification when exposed to an ambient humidity within the humidity range specified in MIL-STD-810 (see 4.8.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

3.8.3 Vibration. The monitor shall withstand vibration tests in accordance with MIL-STD-167-1 without mechanical damage or malfunctioning (see 4.8 7).

3.8.4 Shock. The monitor shall withstand the shock tests specified in MIL-S-901 on the lightweight machine without the use of shock mounts. With rated voltage applied, all monitors shall function properly after the shock test. Mounting fixture 4A of MIL-S-901 shall apply (see 4.8.8).

3.8.5 Electromagnetic interference (EMI). The monitor shall fully meet the EMI requirements of MIL-STD-461 for class II B equipment (see 4.8.9).

3.8.6 Noise. The monitor shall meet the requirements of MIL-STD-740 for type 3 and grade C equipment (see 4.8.10).

3 8.7 Warm-up. The monitor shall meet the performance requirements of this specification after being energized for 1 minute (see 4 8 11)

3.8.8 Dielectric strength. Monitors shall be constructed to withstand for a period of 1 minute a dielectric test voltage between separate circuits and to ground as follows (see 4.8 17).

Input or output voltage (ac rms)	Test voltage (rms)
60 and under	600
61 to 90	900
91 to 600	1000 plus twice rated voltage
Over 600	2000 plus 2-1/4 times rated voltage

3.8.9 Insulation resistance The insulation resistance of all circuits shall be no less than 10 megaohms at 25°C ambient temperature (see 4.8.2).

3.9 Weight and dimensions The weight and dimensions of the monitor shall be as follows:

3.9.1 Weight. The weight of the FM61 monitor shall not exceed 7 pounds (see 4 8.12) The weight of the FM60 monitor shall not exceed 5 pounds.

3 9.2 Dimensions. The dimensions of the FM61 monitor shall correspond to those shown on figure 1 (see 4.8 13). The size of FM60 monitor shall have a maximum volume of 370 cubic inches

3.10 Painting. Parts not fabricated of a corrosion-resisting material or having a corrosion-restistant treatment in accordance with MIL-E-917, except for semi-conductors, shall be painted in accordance with MIL-E-917. Only one coat of gray enamel shall be applied on the outside of the enclosure. Touching-up will be permitted for marks or scratches due to factory handling.

3 11 Identification, marking, and labeling. Label plates shall be in accordance with MIL-E-917

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.

3.11.1 Identification plate. Each monitor shall be provided with an identification plate in accordance with MIL-P-15024 and MIL-P-15024/5 which shall be fastened to the front of the enclosure. The plate shall be brass, corrosion-resistant steel, or plastic not less than 0.030 inch in thickness and marked by etching, engraving, stamping or photoprocess. The marking shall be not less than 0.003 inch deep. The identification plate shall indicate the following information:

FREQUENCY MONITOR

TYPE: (See 1 2) VOLTS: 450 ac FREQUENCY: 60 hertz PHASE: 1

FREQ TRIP BAND: (See 3.4.2(d)) DELAY: NONE

TRIP CONTACT: 5A/450V/25W

CAT NO. _____ SER NO _____ YEAR _____

STK. NO. _____ CONTRACT _____ MANUAL: NAVSEA _____

MANUFACTURER'S NAME AND ADDRESS ACCEPTANCE STAMP

3 11.2 Connection diagram plate A wiring diagram similar to figure 2 shall be mounted on the enclosure front. The material and engraving shall be as specified in 3.11.1.

3.12 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

3 13 Workmanship Metal surfaces shall have a smooth finish and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with the best practice for high quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between connections, ruggedness, and suitability of enclosure.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4 3)
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2.1), first article inspection shall be performed on one monitor at its rated frequency. The inspection shall consist of the examination and tests specified in table I.

TABLE I First article and quality conformance inspection

Inspection	Requirement	Test	First article	Quality conformance
General examination	3 4	4.7.1	X	X
Effectiveness of enclosure	3 4.3	4.8 1	X	
Insulation resistance	3 8 9	4 8.2	X	
Leakage current	3 2.2	4.8 3	X	
Input power	3.4.4	4 8 4	X	
Temperature	3 8.1	4.8 5	X	
Humidity	3 8.2	4 8 6	X	
Vibration	3.8 3	4.8 7	X	
Shock	3.8.4	4.8.8	X	
EMI	3.8 5	4.8.9	X	
Noise	3.8.6	4.8 10	X	
Warm-up	3.8.7	4.8.11	X	
Weight and dimensions	3 9	4.8.12, 4 8.13	X	
Performance/sensing	3.7	4.8 14	X	X
Duty	3.4 8	4.8 14 3	X	
Reliability	3.5	4.8.15	X	
Maintainability	3.6	4.8.16	X	
Dielectric strength	3.8 8	4.8.17	X	X

4.3.1 First article inspection report. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2 2).

4.4 Quality conformance inspections. Quality conformance inspections shall be as specified in table I. Unless otherwise specified (see 6.2.1), all inspections and tests shall be conducted in the order listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

4.4 1 Sampling for quality conformance inspection. Quality conformance sampling shall be as follows:

4.4.1 1 Inspection lot. Inspection lot shall be in accordance with MIL-STD-105 level II. Acceptable quality level (AQL) shall be 0 01. Equipment of the same type and size identified for inspection at one time shall be considered a lot for quality conformance inspection.

4 4.2 Classification of defects. Failure to satisfy any test of 4.8 shall be counted as a defect. Unless otherwise approved by the contracting activity, all defects are classified as major.

4.4.3 Resubmittal of lots. Resubmittal of rejected lots shall be in accordance with MIL-STD-105.

4 4.4 Quality conformance inspection report. When specified in the contract or order, a quality conformance inspection report shall be prepared (see 6 2 2)

4 5 Test equipment. Test equipment shall be as required in MIL-E-917 except that the shock machine shall be in accordance with MIL-S-901.

4.6 Test conditions Unless otherwise required in the detailed test herein, the tests of table I shall be performed in accordance with the test conditions of MIL-STD-202, except as follows. Ambient conditions within the specified ranges need not be controlled.

(a) Temperature: FM60 - 0 - 65°C
FM61 - 0 - 75°C
(b) Humidity: 0 to 70 percent relative humidity
(c) Power input: Nominal power of 3.4.5
(d) Attitude· Normal operating position

4.7 Detailed inspection. Detailed inspection shall be as follows:

4 7 1 General examination. The completed monitor shall be given a thorough examination which will include the following.

(a) Workmanship, assembly, and fit.
(b) Parts, materials, color, and finishes.
(c) Treatment for prevention of corrosion.
(d) Markings correct for the specific application and correctly placed.
(e) Conformance to safety requirements of 3.2.
(f) Proper harness dress.
(g) Labeling and identification.
(h) Ascertaining that all parts, including repair parts, are accounted for in the final deliverable package.

4.8 Tests. Tests shall be conducted in accordance with 4.8.1 through 4.8.17

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

4.8.1 Effectiveness of enclosure. The FM60 and FM61 monitor enclosures shall be tested in accordance with MIL-E-2036 for dripproof and totally enclosed requirements, respectively. Enclosures shall be accepted or rejected based on inspection to ascertain that each enclosure performs its specified function (see 3.4.3).

4.8.2 Insulation resistance. Insulation resistance shall be as specified in 3 8 9. The measurement of insulation resistance shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. Insulation resistance shall be measured with an insulation-resistance-indicating meter at 500 volts dc. The time of test voltage application shall be not less than 60 seconds. Measurements shall be made at any convenient ambient temperature with all circuits cold. The relative humidity and ambient temperature shall be recorded. The monitor shall not be damaged when the generator cables or the external monitor leads are tested between phases with the 500 volt dc tester (see 3.8.9)

4.8 3 Leakage current The leakage current test shall be performed on monitors constructed for permanent installation. Test methods and leakage current limits shall be in accordance with MIL-E-917 (see 3 2 2)

4 8 4 Input power. Input power tests shall be in accordance with DOD-STD-1399, section 300 (see 3 4.5).

4.8.5 Temperature test. This test shall be performed at 0°C and maximum ambient temperature. The device shall be held for 168 hours at each temperature The voltage shall be varied from 410 to 470 volts and operated once a day in accordance with 4 8.14 Each day, the voltage shall be raised to 510 volts for 2 seconds. Temperature rise of the case shall not exceed 15°C (see 3.8.1)

4.8.6 Humidity Humidity tests shall be in accordance with method 507.2, procedure III of MIL-STD-810, except as follows (see 3.8.2).

Step 5 - Monitor performance measurements shall be made at plus 50°C for the FM60 and at plus 65°C for the FM61.
Step 9 - The monitor shall be carefully examined in detail to detect evidence of failure. Blistering of organic coatings shall be cause for rejection

4.8.7 Vibration The monitor shall be subjected to type I vibration tests in accordance with MIL-STD-167-1 when energized at rated voltage and frequency (see 3 8 3).

4 8 8 Shock Shock tests shall be conducted in accordance with MIL-S-901 for grade A, class I equipment (see 3 8 4). After the shock tests, the monitor shall be given an operational test for conformance to 3 7.

4.8 9 Electromagnetic interference. The monitor shall be tested in accordance with MIL-STD-462 (see 3.8 5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:08Z
Check the source to verify that this is the current version before use.

4.8.10 Noise. Noise tests shall be conducted in accordance with MIL-STD-740 for type 3, grade C equipment (see 3.8 6). The monitor shall be energized simulating actual operation.

4.8.11 Warm-up. When the monitor is subjected to rated voltage and frequency at both 20 and 65°C ambient, the equipment shall meet the requirements of 3.8.7.

4.8.12 Weight. The monitors shall be weighed to demonstrate conformance to 3.9.1.

4.8.13 Dimensions. The FM61 monitor shall be measured to demonstrate conformance to figure 1. The size of FM60 monitor shall be as specified in 3.9.2.

4.8.14 Performance/sensing. The performance of the monitor shall be verified by connecting the unit as shown on figure 2 The test circuit shall be supplied from a variable frequency generator and shall consist of a relay and inductive circuit of 25 watts at 0 15. The test circuit hereafter is referred to as the circuit breaker. The test coil shall be continuously energized in the trip condition (see 3 7)

4 8.14.1 Test condition number 1. The test circuit condition shall be a nominal 440 volts ac, unless the test is being performed in conjunction with another test. The test procedure shall be a follows:

(a) Suddenly disconnect the drive to the generator and record the voltage and frequency values at which the monitor caused the breaker to open. The monitor shall trip the breaker within tolerance of setting. This shall be repeated five times.
(b) Slowly reduce the speed of the generator and record the voltage and frequency values at which the monitor caused the breaker to open The monitor shall open the breaker within tolerance of setting. This shall be repeated five times.
(c) Try to close the breaker below tolerance of setting and at a maximum voltage of 270 volts. The breaker shall not close.
(d) Try to close the breaker at high tolerance of setting and 370 volts. The breaker shall close. This shall be repeated five times.
(e) Disconnect one sensing lead The breaker shall open. Record.
(f) With monitor bypass switch open, operate generator over a range of 30 to 60 Hz. The breaker shall not open.

4.8.14 2 Test condition number 2. Raise the voltage and frequency slowly. The breaker shall close (relay deenergized) below 374 volts and within tolerance of frequency setting Repeat five times For the FM61, this shall be done at high, mid and low settings.

4 8.14.3 Test condition number 3 The unit shall be operated at nominal voltage by lowering and raising the frequency 40 times in 10 minutes. The monitor shall trip the circuit breaker and reset itself as the frequency passes through the set range.

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.

4.8.15 Reliability. Reliability tests shall be as follows·

4.8.15.1 Demonstration phase. Unless otherwise specified (see 6.2 1), the demonstration phase of reliability testing shall be conducted on at least five (but not more than 10) monitors in accordance with test plan IIIC of MIL-STD-781. The upper test MTBF shall be as specified in 3.5.

4.8.15.2 Test schedule. The test shall consist of an uninterrupted series of three 8-hour periods per day. One of the three periods shall be manned The remaining two periods need not be manned.

4.8.15.3 Stress conditions. Unless otherwise specified below or in the contract or order (see 6.2.1), environmental and electrical stress conditions shall be as specified under the combined environments for shipboard equipment (sheltered) in appendix B of MIL-STD-781.

(a) Electrical stress. When operation from two or more nominal voltages is required, the nominal voltage for reliability testing shall be the highest required nominal voltage
(b) Thermal stress: The test time for each thermal cycle shall be 8 hours

4.8.15.4 Sampling phase. When specified (see 6.2.1), the sampling phase of reliability testing shall be conducted on at least five (but not more than 10) monitors in accordance with test plan IVC of MIL-STD-781. The upper test MTBF shall be as specified in 3.5 The test schedule and stress conditions of 4.8.15.2 and 4.8 15.3 shall apply.

4.8.16 Maintainability demonstration. When specified (see 6 2.1), conformance to 3.6 shall be verified through a maintenance demonstration procedure, maintenance task selection, and maintenance task performance in accordance with MIL-STD-471.

4.8 17 Dielectric strength tests Dielectric strength tests shall be as follows:

4.8 17.1 General. The dielectric strength test shall be made after all of the tests specified in 4.7 through 4 8.16 have been completed The dielectric tests shall be made on the completely assembled monitor and not on individual parts.

4.8 17.2 Dielectric voltage test. The frequency of the testing voltage shall be not less than 60 Hz and shall approximate a true sine wave. The value of test voltage shall be as specified in 3.8 8 and shall be applied continuously for a period of 1 minute Rectifiers, bridges, and zener references shall be shorted out before applying the voltage.

4.8 17.3 Measurement of test voltage The measurement of the voltage used in dielectric tests shall be made by the voltmeter method whereby the meter drives its voltage from the high-voltage circuit either directly, by means of a voltmeter coil placed in the testing transformer, or through an auxiliary ratio transformer.

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the currentversion before use.

4 8 17 4 Points of application The test voltage shall be successively applied between each electric circuit and all other electric circuits and grounded metal parts not electrically connected to it.

4 8 18 Nonconformance. The monitor shall be rejected if it fails any one of the following:

(a) Breakage or appreciable distortions of any parts.
(b) Electrical malfunction. The sensing features shall not malfunction, become inoperative, or deviate from specifications.
(c) Insulation resistance shall not vary from its original value.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), frequency monitors shall be preserved level A, C or commercial, packed level A, B, C or commercial and marked in accordance with MIL-E-17555.

6. NOTES

6 1 Intended use. The monitor should be connected to the output terminals of rotating units supplying power (see figure 2). Upon sensing unsuitable frequency conditions, the monitor should cause the associated circuit breaker to open

6.2 Ordering data

6 2.1 Acquisition requirements Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of monitor required (see 1.2).
(c) Whether first article is required (see 3 1 and 4.3).
(d) Materials, parts, and processes to be used, if other than specified herein (see 3.3).
(e) Whether use of a printed circuit board is permitted (see 3.4.6).
(f) Operating voltage (see 3.4.9).
(g) Operating frequency and type of power (see 3.4.10).
(h) Environmental requirements, if other than specified herein (see 3.8).
(i) Whether quality assurance tests must be performed in the order specified herein (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.

(j) Conditions for reliability tests, if other than specified herein (see 4.8.15).
(k) Whether maintainability demonstration is required (see 4.8.16).
(l) Preservation, packaging, packing, and marking, if other than specified (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.12	Drawings, engineering and associated lists	DI-E-7031	Level 2
4.3 1	First article inspection report	DI-T-4902	----
4.4.4	Reports, test	DI-T-2072	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010 12-L , AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article When a first article inspection is required, the item should be a first article sample The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6 6 Definitions. The definitions of Section 300 of DOD-STD-1399 and MIL-STD-109 are applicable herein.

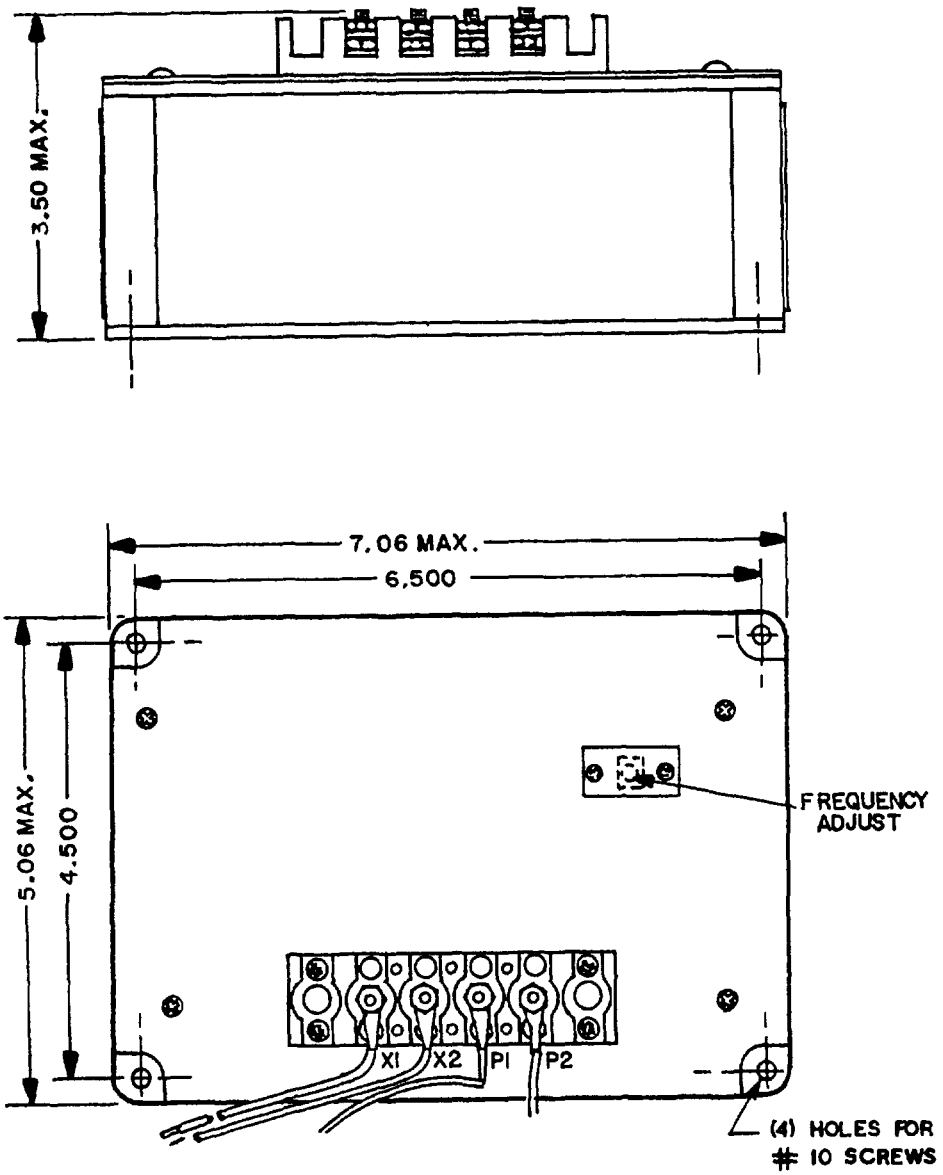
6.7 Subject term (key word) listing.

Circuit breaker
Frequency value
Monitor

6 8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity.
Navy - SH
(Project 5925-N099)

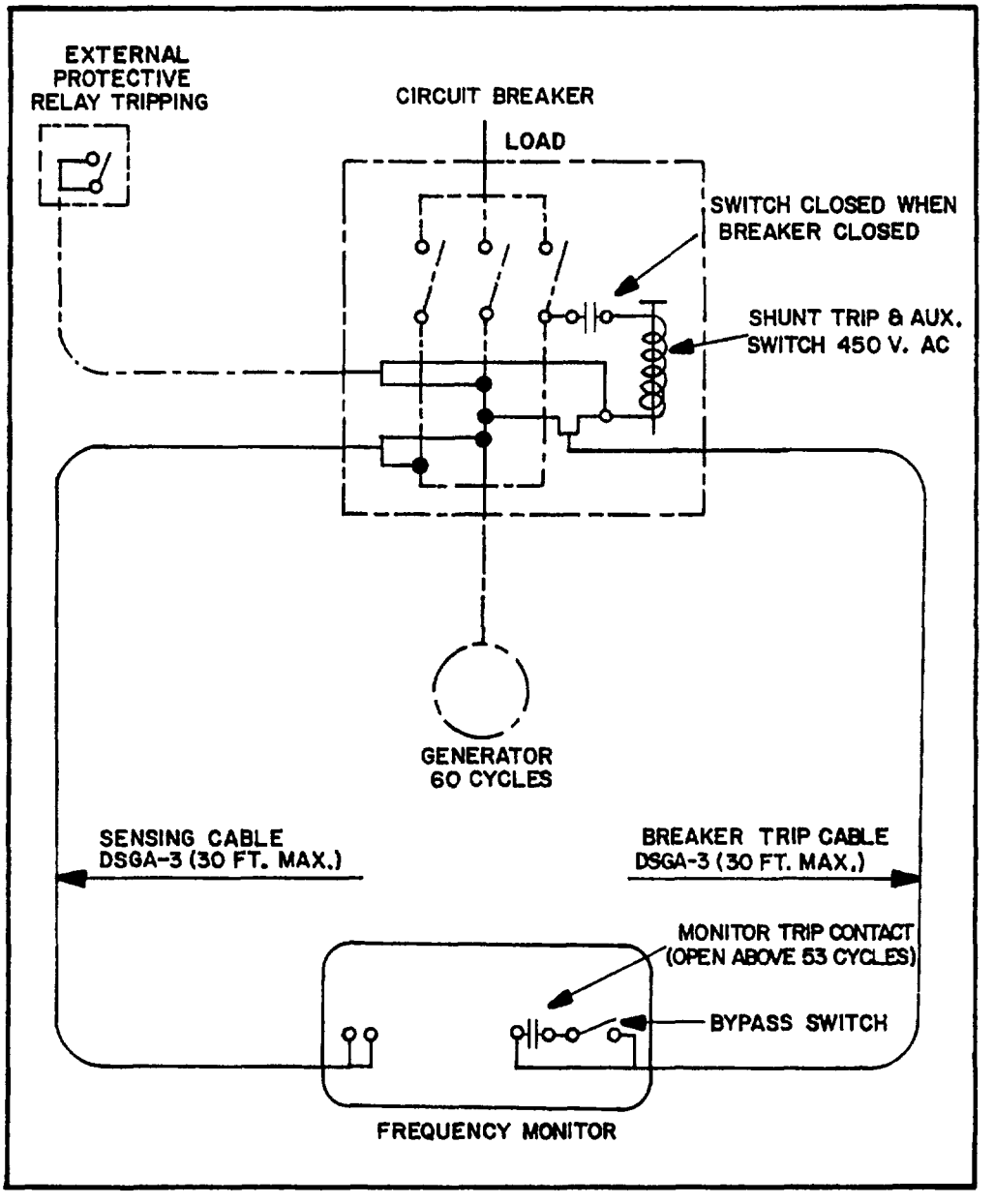
Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.



SH 131765

FIGURE 1 Outline and mounting dimensions for FM61 monitor.

Source: https://assist.dla.mil -- Downloaded: 20260-19T20:08Z
Check the source to verify that this is the current version before use.



SH 131871

FIGURE 2. Typical external circuit connection for FM60 and FM61 monitor.

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along thu line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-24225A(SH)	MONITOR, FREQUENCY, 60 HERTZ, ELECTRICAL (NAVAL SHIPBOARD USE)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202640-19T20:08Z
Check the source to verify that this is the current version before use.